Year 9

REVISION PACK

Unit 2

ANSWERS

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Unit 2 · Prisms (Lessons 2.1, 2.2)

Q	Working	Answer
triangular prisms		
1.1	**(a)** $\frac{1}{2} \times 12 \times 5 = 30$ $30 \times 20 = 600$ **(b)** $2 \times 30 = 60$ $12 \times 20 + 5 \times 20 + 13 \times 20 = 600$ $60 + 600 = 660$	**(a) 600 cm³** **(b) 660 cm²**
1.2	**(a)** $\frac{1}{2} \times 12 \times 9 = 54$ $54 \times 10 = 540$ **(b)** $2 \times 54 = 108$ $12 \times 10 + 9 \times 10 + 15 \times 10 = 360$ $108 + 360 = 468$	**(a) 540 cm³** **(b) 468 cm²**
1.3	**(a)** $\frac{1}{2} \times 15 \times 8 = 60$ $60 \times 12 = 720$ **(b)** $2 \times 60 = 120$ $15 \times 12 + 8 \times 12 + 17 \times 12 = 480$ $120 + 480 = 600$	**(a) 720 cm³** **(b) 600 cm²**
1.4	**(a)** $\frac{1}{2} \times 20 \times 21 = 210$ $210 \times 10 = 2100$ **(b)** $2 \times 210 = 420$ $20 \times 10 + 21 \times 10 + 29 \times 10 = 700$ $420 + 700 = 1120$	**(a) 2100 cm³** **(b) 1120 cm²**
1.5	**(a)** $\frac{1}{2} \times 9 \times 12 = 54$ $54 \times 20 = 1080$ **(b)** $2 \times 54 = 108$ $9 \times 20 + 12 \times 20 + 15 \times 20 = 720$ $108 + 720 = 828$	**(a) 1080 cm³** **(b) 828 cm²**

1.6	**(a)** $\frac{1}{2} \times 5 \times 12 = 30$ $30 \times 8 = 240$ **(b)** $2 \times 30 = 60$ $5 \times 8 + 12 \times 8 + 13 \times 8 = 240$ $60 + 240 = 300$	**(a) 240 cm³** **(b) 300 cm²**
litres in a tank		
1.7	$60 \times 25 \times 40 = 60\,000$ $60\,000 \div 1000 = 60$	**60 litres**
1.8	$80 \times 35 \times 30 = 84\,000$ $84\,000 \div 1000 = 84$	**84 litres**
1.9	$45 \times 20 \times 50 = 45\,000$ $45\,000 \div 1000 = 45$	**45 litres**
1.10	$70 \times 30 \times 60 = 126\,000$ $126\,000 \div 1000 = 126$	**126 litres**
1.11	$25 \times 24 \times 50 = 30\,000$ $30\,000 \div 1000 = 30$	**30 litres**
1.12	$90 \times 40 \times 25 = 90\,000$ $90\,000 \div 1000 = 90$	**90 litres**

Unit 2 · Circles (Lessons 2.3, 2.4)

Q	Working	Answer
circumference		
2.1	$2 \times \pi \times 9 = 56.54867\ldots$	**56.5 cm**
2.2	$2 \times \pi \times 4.5 = 28.27433\ldots$	**28.3 cm**
2.3	$2 \times \pi \times 11 = 69.11504\ldots$	**69.1 cm**
2.4	$2 \times \pi \times 6.5 = 40.8407\ldots$	**40.8 cm**
2.5	$2 \times \pi \times 13 = 81.68141\ldots$	**81.7 cm**
2.6	$2 \times \pi \times 3.5 = 21.99115\ldots$	**22.0 cm**

area of a circle		
2.7	$r = 14 \div 2 = 7$ $\pi \times 7^2 = 153.938\ldots$	**153.9 cm²**
2.8	$r = 6 \div 2 = 3$ $\pi \times 3^2 = 28.27433\ldots$	**28.3 cm²**
2.9	$r = 12 \div 2 = 6$ $\pi \times 6^2 = 113.0973\ldots$	**113.1 cm²**
2.10	$r = 18 \div 2 = 9$ $\pi \times 9^2 = 254.469\ldots$	**254.5 cm²**
2.11	$r = 9 \div 2 = 4.5$ $\pi \times 4.5^2 = 63.61725\ldots$	**63.6 cm²**
2.12	$r = 20 \div 2 = 10$ $\pi \times 10^2 = 314.1593\ldots$	**314.2 cm²**
perimeter of a semicircle		
2.13	$\pi \times 16 \div 2 = 25.13274\ldots$ $25.13274\ldots + 16 = 41.13274\ldots$	**41.1 cm**
2.14	$\pi \times 9 \div 2 = 14.13717\ldots$ $14.13717\ldots + 9 = 23.13717\ldots$	**23.1 cm**
2.15	$\pi \times 20 \div 2 = 31.41593\ldots$ $31.41593\ldots + 20 = 51.41593\ldots$	**51.4 cm**
2.16	$\pi \times 14 \div 2 = 21.99115\ldots$ $21.99115\ldots + 14 = 35.99115\ldots$	**36.0 cm**
2.17	$\pi \times 7 \div 2 = 10.99557\ldots$ $10.99557\ldots + 7 = 17.99557\ldots$	**18.0 cm**
2.18	$\pi \times 18 \div 2 = 28.27433\ldots$ $28.27433\ldots + 18 = 46.27433\ldots$	**46.3 cm**
radius from the area		

2.19	$\pi r^2 = 40$ $r^2 = 40 \div \pi = 12.7324\ldots$ $r = \sqrt{12.7324\ldots} = 3.568248\ldots$	**3.6 cm**
2.20	$\pi r^2 = 120$ $r^2 = 120 \div \pi = 38.19719\ldots$ $r = \sqrt{38.19719\ldots} = 6.180387\ldots$	**6.2 cm**
2.21	$\pi r^2 = 65$ $r^2 = 65 \div \pi = 20.69014\ldots$ $r = \sqrt{20.69014\ldots} = 4.548642\ldots$	**4.5 cm**
2.22	$\pi r^2 = 30$ $r^2 = 30 \div \pi = 9.549297\ldots$ $r = \sqrt{9.549297\ldots} = 3.090194\ldots$	**3.1 cm**
2.23	$\pi r^2 = 200$ $r^2 = 200 \div \pi = 63.66198\ldots$ $r = \sqrt{63.66198\ldots} = 7.978846\ldots$	**8.0 cm**
2.24	$\pi r^2 = 90$ $r^2 = 90 \div \pi = 28.64789\ldots$ $r = \sqrt{28.64789\ldots} = 5.352372\ldots$	**5.4 cm**

Unit 2 · Cylinders (Lesson 2.5)

Q	Working	Answer
volume and surface area of a cylinder		
3.1	**(a)** $\pi \times 5^2 \times 12 = 942.4778\ldots$ **(b)** $2 \times \pi \times 5^2 = 157.0796\ldots$ $2 \times \pi \times 5 \times 12 = 376.9911\ldots$ $157.0796\ldots + 376.9911\ldots = 534.0708\ldots$	**(a) 942.5 cm³** **(b) 534.1 cm²**
3.2	**(a)** $\pi \times 2^2 \times 7 = 87.96459\ldots$ **(b)** $2 \times \pi \times 2^2 = 25.13274\ldots$ $2 \times \pi \times 2 \times 7 = 87.96459\ldots$ $25.13274\ldots + 87.96459\ldots = 113.0973\ldots$	**(a) 88.0 cm³** **(b) 113.1 cm²**

3.3	**(a)** $\pi \times 6^2 \times 8 = 904.7787\ldots$ **(b)** $2 \times \pi \times 6^2 = 226.1947\ldots$ $2 \times \pi \times 6 \times 8 = 301.5929\ldots$ $226.1947\ldots + 301.5929\ldots = 527.7876\ldots$	**(a) 904.8 cm³** **(b) 527.8 cm²**
3.4	**(a)** $\pi \times 3^2 \times 11 = 311.0177\ldots$ **(b)** $2 \times \pi \times 3^2 = 56.54867\ldots$ $2 \times \pi \times 3 \times 11 = 207.3451\ldots$ $56.54867\ldots + 207.3451\ldots = 263.8938\ldots$	**(a) 311.0 cm³** **(b) 263.9 cm²**
3.5	**(a)** $\pi \times 7^2 \times 5 = 769.6902$ **(b)** $2 \times \pi \times 7^2 = 307.8761\ldots$ $2 \times \pi \times 7 \times 5 = 219.9115\ldots$ $307.8761\ldots + 219.9115\ldots = 527.7876\ldots$	**(a) 769.7 cm³** **(b) 527.8 cm²**
3.6	**(a)** $\pi \times 4.5^2 \times 10 = 636.1725\ldots$ **(b)** $2 \times \pi \times 4.5^2 = 127.2345\ldots$ $2 \times \pi \times 4.5 \times 10 = 282.7433\ldots$ $127.2345\ldots + 282.7433\ldots = 409.9778\ldots$	**(a) 636.2 cm³** **(b) 410.0 cm²**

Unit 2 · Pythagoras' theorem (Lesson 2.6)

Q	Working	Answer
finding the hypotenuse		
4.1	$x^2 = 5^2 + 12^2 = 169$ $x = \sqrt{169} = 13$	x **= 13 cm**
4.2	$x^2 = 7^2 + 24^2 = 625$ $x = \sqrt{625} = 25$	x **= 25 cm**
4.3	$x^2 = 6^2 + 9^2 = 117$ $x = \sqrt{117} = 10.81665\ldots$	x **= 10.8 cm**
4.4	$x^2 = 9^2 + 40^2 = 1681$ $x = \sqrt{1681} = 41$	x **= 41 cm**

4.5	$x^2 = 4^2 + 7^2 = 65$ $x = \sqrt{65} = 8.062258\ldots$	x **= 8.1 cm**
4.6	$x^2 = 11^2 + 13^2 = 290$ $x = \sqrt{290} = 17.02939\ldots$	x **= 17.0 cm**
finding a shorter side		
4.7	$y^2 = 10^2 - 6^2 = 64$ $y = \sqrt{64} = 8$	y **= 8 cm**
4.8	$y^2 = 25^2 - 7^2 = 576$ $y = \sqrt{576} = 24$	y **= 24 cm**
4.9	$y^2 = 12^2 - 5^2 = 119$ $y = \sqrt{119} = 10.90871\ldots$	y **= 10.9 cm**
4.10	$y^2 = 26^2 - 10^2 = 576$ $y = \sqrt{576} = 24$	y **= 24 cm**
4.11	$y^2 = 20^2 - 16^2 = 144$ $y = \sqrt{144} = 12$	y **= 12 cm**
4.12	$y^2 = 9^2 - 4^2 = 65$ $y = \sqrt{65} = 8.062258\ldots$	y **= 8.1 cm**
the diagonal of a rectangle		
4.13	$d^2 = 8^2 + 5^2 = 89$ $d = \sqrt{89} = 9.433981\ldots$	**9.4 cm**
4.14	$d^2 = 12^2 + 7^2 = 193$ $d = \sqrt{193} = 13.89244\ldots$	**13.9 cm**
4.15	$d^2 = 10^2 + 3^2 = 109$ $d = \sqrt{109} = 10.44031\ldots$	**10.4 cm**
4.16	$d^2 = 7^2 + 2^2 = 53$ $d = \sqrt{53} = 7.28011\ldots$	**7.3 cm**
4.17	$d^2 = 11^2 + 6^2 = 157$ $d = \sqrt{157} = 12.52996\ldots$	**12.5 cm**

4.18	$d^2 = 15^2 + 4^2 = 241$ $d = \sqrt{241} = 15.52417\ldots$	**15.5 cm**

Unit 2 · Units of volume (Lesson 2.2)

Q	Working	Answer
litres into cm³		
5.1	$4.2 \times 1000 = 4200$	**4200 cm³**
5.2	$0.75 \times 1000 = 750$	**750 cm³**
5.3	$6 \times 1000 = 6000$	**6000 cm³**
5.4	$1.35 \times 1000 = 1350$	**1350 cm³**
5.5	$0.4 \times 1000 = 400$	**400 cm³**
5.6	$12.5 \times 1000 = 12\,500$	**12 500 cm³**